Unit: 6 Unit 7: Factoring
Topic: Sum and Difference of Cubes

Name: ________________
Date: ___________

Advanced (Habanero)

Q1. What is the factored form of $x^3 + y^3$?
(A) $(x + y)(x^2 - xy + y^2)$
(B) $(x - y)(x^2 + xy + y^2)$
(C) $(x + y)(x^2 + xy - y^2)$
(D) $(x - y)(x^2 - xy - y^2)$
(E) $(x + y)(x^2 - xy - y^2)$

Q2. What is the factored form of $8x^3 - 27$?
(A) $(2x - 3)(4x^2 + 6x + 9)$
(B) $(2x + 3)(4x^2 - 6x + 9)$
(C) $(2x - 3)(4x^2 - 6x - 9)$
(D) $(2x + 3)(4x^2 + 6x - 9)$
(E) $(2x - 3)(4x^2 + 6x - 9)$

Q3. What is the factored form of $x^3 - 8y^3$?
(A) $(x - 2y)(x^2 + 2xy + 4y^2)$
(B) $(x + 2y)(x^2 - 2xy + 4y^2)$
(C) $(x - 2y)(x^2 - 2xy + 4y^2)$
(D) $(x + 2y)(x^2 + 2xy + 4y^2)$
(E) $(x - 2y)(x^2 + 2xy - 4y^2)$

Q4. What is the factored form of $27x^3 + 8y^3$?
(A) $(3x + y)(9x^2 - 3xy + y^2)$
(B) $(3x - y)(9x^2 + 3xy + y^2)$
(C) $(3x + y)(9x^2 + 3xy - y^2)$
(D) $(3x - y)(9x^2 - 3xy - y^2)$
(E) $(3x + y)(9x^2 - 3xy - y^2)$

Q5. What is the factored form of $x^3 + 64$?
(A) $(x + 4)(x^2 - 4x + 16)$
(B) $(x - 4)(x^2 + 4x + 16)$
(C) $(x + 4)(x^2 + 4x - 16)$
(D) $(x - 4)(x^2 - 4x - 16)$
(E) $(x + 4)(x^2 - 4x - 16)$

Q6. What is the factored form of $125x^3 - 8y^3$?
(A) $(5x - 2y)(25x^2 + 10xy + 4y^2)$
(B) $(5x + 2y)(25x^2 - 10xy + 4y^2)$
(C) $(5x - 2y)(25x^2 - 10xy + 4y^2)$
(D) $(5x + 2y)(25x^2 + 10xy + 4y^2)$
(E) $(5x - 2y)(25x^2 + 10xy - 4y^2)$

Q7. What is the factored form of $64x^3 + 27y^3$?
(A) $(4x + 3y)(16x^2 - 12xy + 9y^2)$
(B) $(4x - 3y)(16x^2 + 12xy + 9y^2)$
(C) $(4x + 3y)(16x^2 + 12xy - 9y^2)$
(D) $(4x - 3y)(16x^2 - 12xy - 9y^2)$
(E) $(4x + 3y)(16x^2 - 12xy - 9y^2)$

Q8. What is the factored form of $8x^3 - y^3$?
(A) $(2x - y)(4x^2 + 2xy + y^2)$
(B) $(2x + y)(4x^2 - 2xy + y^2)$
(C) $(2x - y)(4x^2 - 2xy + y^2)$
(D) $(2x + y)(4x^2 + 2xy + y^2)$
(E) $(2x - y)(4x^2 + 2xy - y^2)$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^3 + 125y^3$.

Q10. Factor the expression $64x^3 - 27y^3$.

Chef's Tips

- Make sure students understand the difference between sum and difference of cubes.
- Emphasize the importance of factoring out the greatest common factor before applying the sum or difference of cubes formula.
- Encourage students to use the formulas $(a + b)(a^2 - ab + b^2)$ and $(a - b)(a^2 + ab + b^2)$ to factor expressions of the form a^3pmb^3.